

PRANAVI JAIN

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SUMMARY

Quantum Software Engineer with experience building modular Python frameworks for quantum experimentation, data pipelines, and circuit analysis across simulators and cloud hardware backends. Background in enterprise backend services, API development, and production backend systems. Seeking to contribute to quantum data infrastructure and software tooling at Google Quantum AI.

EDUCATION

University of Southern California (USC)

M.S. Quantum Information Science | GPA – 3.57/4

Los Angeles, CA

August 2023 – May 2025

Thapar Institute of Engineering and Technology

B.E. Electronics and Computers | CGPA – 9.03/10

Patiala, India

July 2017 – June 2021

WORK EXPERIENCE

Quantum Software Engineer – NeoQuortex

College Park, MD

June 2025 – Present

- Build and maintain a **modular Python framework** for hybrid quantum-classical experiments, with execution against quantum simulators and cloud backends.
- Designed **abstraction layers** separating algorithm logic, circuit construction, and backend execution — providing a clean programmatic interface for experimentation across backends without modifying core framework logic.
- Implemented **structured logging** and **version control practices** to ensure reproducibility and maintainability of results data.
- Built **orchestration pipelines** for structured batch execution and reproducible benchmarks, managing experiment metadata, parameter sets, and result datasets.
- Built benchmarking pipelines comparing parametrized quantum circuits across multiple algorithms, producing **comparative performance datasets** used for internal ansatz selection.

Software Developer – Amdocs

Gurgaon, India

July 2021 – July 2023

- Delivered **Java-based backend services** for telecom OMS/CRM platforms supporting high-revenue customer deployments.
- Led proof-of-concept engagements directly with Singtel and supported **production releases**, including that of Singtel RED.
- Built **SOAP APIs** and **database integrations** for Singtel's digital platforms, handling transactional workflows.
- Authored documentation and conducted **knowledge transfer sessions** for cross-team onboarding.

Web Development Intern – Saathi Global Education Network

Remote, India

January 2021 – June 2021

- Led a 3-person team building a **Django-based CMS platform** with REST APIs and SQL database backend.

ACADEMIC PROJECTS

- **Benchmarking Quantum Computers via Grover's Algorithms** – Executed a small-scale Grover's algorithm across IBM, IonQ, IQM, and Rigetti hardware to analyze circuit resource scaling and cross-platform variation.
- **Zero-Noise Extrapolation (ZNE) for Quantum Error Mitigation** – Implemented ZNE workflow and its variants using Qiskit and **Mitiq** to analyze noise scaling behavior and error mitigation effectiveness.
- **Two-Qubit Entanglement Dynamics on IBM Quantum Systems** – Experimental study of entanglement dynamics on IBM processors by simulating Nakajima-Zwanzig and Time-Convolutionless (TCL) master equations to compare theoretical predictions against hardware noise.

SKILLS

- **Languages:** Python (advanced), Java, C/C++, SQL, Kotlin (working knowledge)
- **Quantum:** PennyLane, Qiskit, AWS BraKet, OpenQASM, Cirq (familiar), Mitiq, NumPy, Pandas
- **Infrastructure:** Linux/Unix, Git/GitHub, GitHub Actions, REST APIs, Soap UI
- **Mathematics:** Linear algebra, numerical optimization, variational algorithms, quantum error correction and mitigation